higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING TECHNOLOGY
NQF LEVEL 2

(6021012)

13 November 2019 (Y-Paper)
13:00–16:00

This question paper consists of 7 pages and 1 addendum.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. ALL sketches must be neat, clear and in proportion.
5. Start each question on a NEW page.
6. Drawings must be done on a clean A4 sheet.
7. Write neatly and legibly.

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SECTION A

QUESTION 1: GENERAL

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

1.1.1 Which measuring instrument would be used for determining the size of the gap in a spark plug?

A Radius gauge
B Thread gauge
C Feeler gauge
D Tape measure

1.1.2 A surface or datum surface is usually used for …

A supporting round shafts for marking.
B accurately marking off pieces of metal.
C making flat surfaces on metal.
D supporting work pieces.

1.1.3 A guillotine is used for …

A removing metal gradually.
B cutting sheet metal.
C removing screws and bolts.
D making scribed lines clearer.

1.1.4 A hydrometer is a tool for measuring …

A the level of a liquid in a glass.
B pressure of liquids.
C density of liquids.
D acid in a battery.

1.1.5 The correct grinding machine for sharpening drill bits, chisels and cutting tools is a/an … grinder.

A pencil
B pedestal
C angle
D bench

(5 × 1) (5)

1.2 Choose a description from COLUMN B that matches a/an item/word in COLUMN A. Write only the letter (A–F) next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.2.1	Tap	A	a tool that is used to measure the diameter of a piston
1.2.2	Divider	B	has a sharp point and is used to draw lines on metal
1.2.3	Outside micrometer	C	used to hold small things and to bend, twist and cut wire
1.2.4	Scriber	D	used to draw circles on a metal plate
1.2.5	Pliers	E	used to cut internal threads in a drilled hole
		F	a flat tool with a straight edge, often shaped like an L

(5 × 1) (5)

1.3 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.3.1–1.3.5) in the ANSWER BOOK.

1.3.1 Tape measures can become rusty and damaged if they are not stored in their proper place.

1.3.2 The correct sizes of drawing sheets are A7, A20 and A30.

1.3.3 Use a piece of wood to stop a grinding machine when the ON/OFF button is not working.

1.3.4 In order to construct an ellipse, the major and minor axes must be known.

1.3.5 Power is the rate at which work is done. (5 × 1) (5)

1.4 Give ONE word or term for each of the following descriptions. Write only the word or term next to the question number (1.4.1–1.4.5) in the ANSWER BOOK.

1.4.1 A six-sided figure

1.4.2 An instrument used to measure and set angles

1.4.3 A gauge that is used to check for the concentricity of a work piece

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1.4.4 The internal diameter of a cylinder

1.4.5 A guided shaft that is used to exert pressure (5 × 1) (5)

[20]

TOTAL SECTION A: 20

SECTION B

QUESTION 2: ENGINEERING HAND TOOLS

2.1 List FIVE engineering hand tools used in the technology field. (5)

2.2 List the FOUR classifications of files. (4)

2.3 The incorrect usage of spanners can have negative consequences.

List FOUR safety rules that should be followed when using a spanner. (4)

2.4 Explain the FIVE reasons that hacksaw blade failure occurs in a workshop. (5)

2.5 State the purpose of a pin punch and give ONE example of where it is used. (2)

[20]

QUESTION 3: ENGINEERING POWER TOOLS

3.1 Power tools are used in the manufacturing, engineering and technology fields.

Draft a table showing SIX power tools and their use in the industry. (6 × 2) (12)

3.2 What is the function of the filter as a compressor component? (2)

3.3 Your foreman asks you to check some power tools for defects. List possible checks for your inspection of the following tools:

3.3.1 Hydraulic press (2)

3.3.2 Compressor (2)

3.3.3 Angle grinder (1)

3.3.4 Portable drilling machine (1)

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3.4 You have to select a grinding disc to grind a weld on a steel gate. There are TWO types of discs available. Give the purpose of each of the following grinding discs:

3.4.1 Metal grinding disc

3.4.2 Masonry disc

(2 × 1) (2)
[22]

QUESTION 4: ENGINEERING MEASURING EQUIPMENT AND SI UNITS

4.1 Numerous engineering precision measuring instruments are used in the manufacturing, engineering and technology fields.

Compile a list of a minimum of FIVE commonly used precision measuring instruments. (5)

4.2 Describe ONE component area or place in industry in which the following measuring instruments could be used:

4.2.1 Vernier calliper

4.2.2 External micrometer

4.2.3 Telescopic gauge

4.2.4 Clock gauge

(4 × 1) (4)

4.3 Add the following times: 12 minutes and 30 seconds plus one hour, 56 minutes and 42 seconds. Your answer must be in minutes and seconds. (5)

4.4 Compile a care and maintenance checklist for a Vernier protractor. (4)

4.5 When using slip gauges, what procedure do you need to follow before wringing of the slip gauges occurs? (5)
[23]

QUESTION 5: BASIC ENGINEERING DRAWING

5.1 FIGURE 1, ADDENDUM A (attached), is an isometric view of a machine part. Draw the following to a scale of 1:1 in first-angle orthographic projection:

5.1.1 Front view (A) (3)

5.1.2 Top view (B) (4)

5.1.3 Left view (C) (3)

5.2 Insert ONE dimension in each of the views. (3)

5.3 Insert the projection symbol in the space below the left view. (2)

[15]

TOTAL SECTION B: 80
GRAND TOTAL: 100

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ADDENDUM A

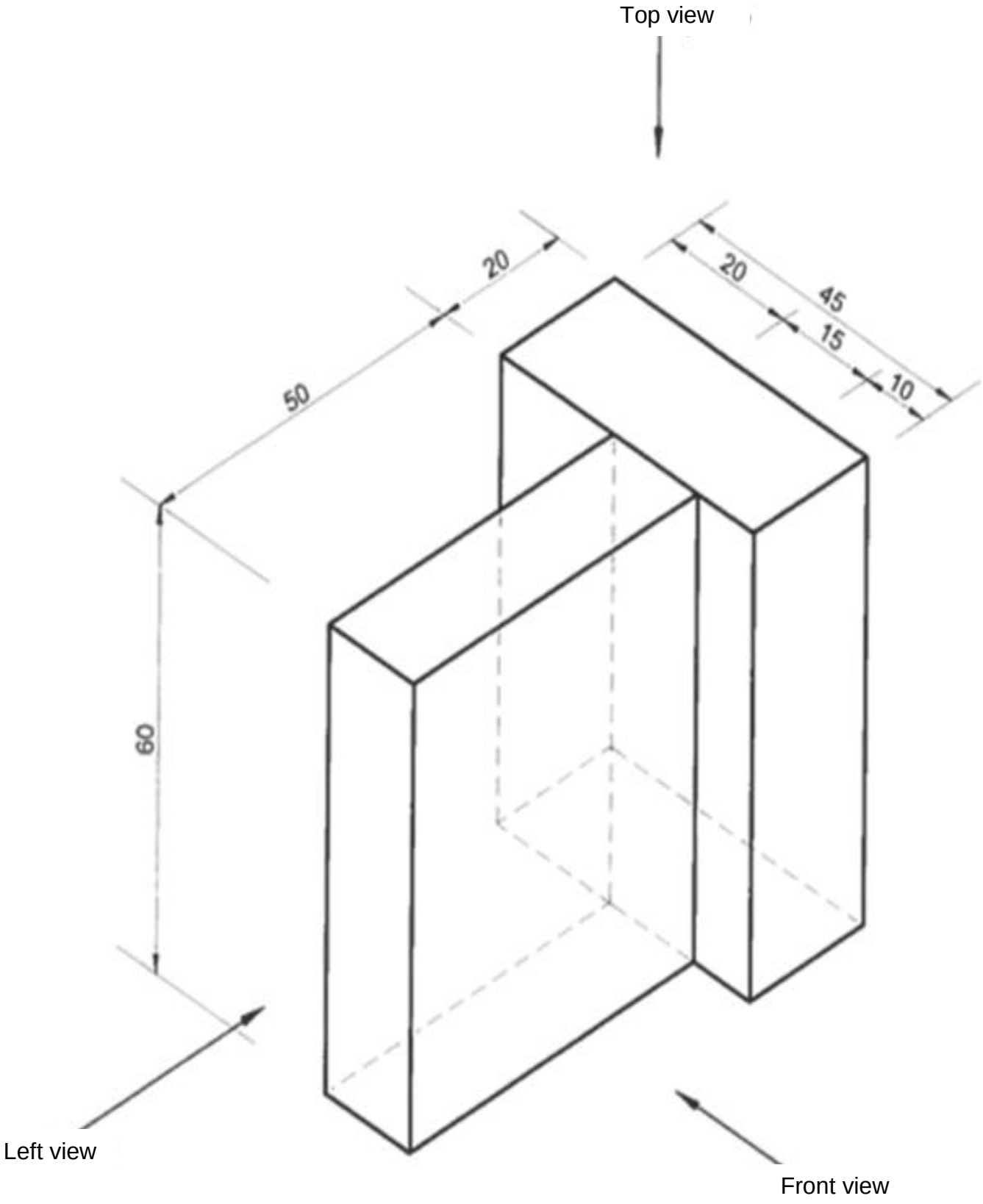


FIGURE 1

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